

Maximum profile likelihood for Cox models under left truncation and right censoring

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Abstract

Co-existence of left truncation and right censoring is common in many survival datasets. To fit a Cox model in this context, a common approach is to still use the partial likelihood method for estimating the regression coefficients, but the risk sets are now determined using both the observed survival times and the left truncation times. The cumulative baseline hazard, however, will then be estimated in a separate step using, for example, the Breslow method. This may cause inference issues if a quantity we wish to infer involves both the baseline hazard and regression coefficients. In this paper, we develop a full likelihood-based method, where when a piecewise constant approximation to the baseline hazard is adopted, a profile log-likelihood can be constructed. The main advantage of our method is that an asymptotic inference procedure for all the model parameters, including the piecewise baseline hazard values, can be developed. Our method also offers fast convergence for computing the piecewise baseline hazard and the Cox regression coefficients. A simulation study comparing our profile likelihood method against the partial likelihood method is conducted and an application to a melanoma data set is reported.

1 Introduction

Left truncation is a common phenomena in many survival (or time-to-event) datasets. Left truncation means any event that occurs before the left truncation time point will

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never be observed (Jin et al., 2022). For instance, an individual may never have the opportunity to participate in a clinical trial if they fail to meet the required conditions for recruitment into the study. A left truncation time is also called a delayed entry time, as it often results from certain conditions imposed on a clinical trial, leading to delayed entries into the study. Ignoring left truncation, meaning treating any left truncation time as time 0 for the follow-up, will lead to biased estimates of the parameters of a survival regression model. It is important to note that there is a fundamental difference between censoring and truncation. Censoring means that a survival time is *partially observed*. In contrast, truncation means that an event time that could have been observed becomes *entirely unobservable* if it falls before the left truncation time.

To fit a Cox model, the conventional method for handling left truncation and right censoring is a modification of the traditional partial likelihood method, where the risk set at any given time consists of individuals whose events or censoring have not yet occurred and whose left truncation times have already passed (e.g., Andersen et al. (1992)). In this approach, Cox model regression coefficients can be computed quickly from maximizing the partial likelihood, for example, using a Newton algorithm. However, a drawback is that the baseline hazard must be estimated separately using an adjusted Breslow method. This separate estimation process will cause inference issues. For example, when making a prediction of a survival probability, the covariance matrix of both the regression coefficients and the baseline hazard estimates is required.

There exist several likelihood-based methods for Cox models with left-truncated survival times, such as Joly et al. (1998), Shen et al. (2019), Shao et al. (2023), and the references therein. A common drawback of these approaches is their high computational cost. Our proposed method in this paper enjoys fast convergence and stable computations.

This paper considers a likelihood-based estimation method. In particular, we propose a profile likelihood approach where the non-parametric baseline hazard is approximated by a piecewise constant function. Under this setting, the solution to the approximate baseline hazard can be easily obtained from its likelihood gradient function. As a result, each piece of the approximate baseline hazard has a closed-form expression involving the Cox regression coefficients. A profile log-likelihood, a function of the regression coefficients only, is then obtained after substituting the piecewise baseline hazard solution into the log-likelihood. Subsequently, this profile log-likelihood is maximized to estimate the regression coefficients, and this process is usually fast and efficient using, for example, the Newton algorithm. A distinctive advantage of this profile likelihood approach, compared to the partial likelihood method, is that it simultaneously estimates both the baseline hazard component and the regression coefficient components of the model, and thus the asymptotic covariance matrix of these estimated parameters can be easily obtained. This makes it feasible to make inferences

on quantities involving both components. Additionally, this approach, similar to the partial likelihood method, enjoys extremely fast convergence speed.

We structure this paper as follows. In Section 2, we first present the likelihood function, and then we derive the profile likelihood. Section 3 explains the Newton algorithm used for efficiently computing the regression coefficient estimates. Some asymptotic results are presented in Section 4. We also conducted a simulation study, with its results reported in Section 5. The application of this approach to a real dataset is included in Section 6. Finally, concluding remarks are given in Section 7.

2 Profile likelihood for left-truncated and right-censored event times

Let T_i denote the event time for individual i , where $i = 1, \dots, n$. We first introduce the notation needed throughout this paper. The observations of individual i are denoted by $(a_i, y_i, \delta_i, \mathbf{x}_i^\top)$. Here, a_i denotes the left truncation time for i , y_i is either the observed T_i or the right censoring time, δ_i represents the event indicator with $\delta_i = 1$ if y_i is an event time and $\delta_i = 0$ otherwise, and $\mathbf{x}_i$ is a vector for the p covariate values of individual i . Note that a_i is strictly less than y_i .

The model we consider in this paper is the conventional semi-parametric Cox model given by

$$h_{T_i}(t|\mathbf{x}_i) = h_0(t)e^{\mathbf{x}_i^\top \boldsymbol{\beta}}, \quad (1)$$

where $h_0(t)$ is an unspecified baseline hazard and $\boldsymbol{\beta}$ denotes a vector for the regression coefficients. We estimate $h_0(t)$ and $\boldsymbol{\beta}$ by maximizing the log-likelihood derived from the observations $(a_i, y_i, \delta_i, \mathbf{x}_i^\top)$ for $i = 1, \dots, n$.

Since a_i is the left truncation time for i , any event for i that occurred before a_i will not be observed. We assume throughout this paper that both a_i and the right censoring time for i are independent of T_i given the covariates, and furthermore, that they are non-informative with respect to the model parameters. Due to truncation, the likelihood function is constructed through the conditional density of T_i given $T_i > a_i$, that is $f_{T_i}(t|T_i > a_i) = f_{T_i}(t)/S_{T_i}(a_i)$, where $f_{T_i}(\cdot)$ and $S_{T_i}(\cdot)$ respectively denote the unconditional density and survival function of T_i . The corresponding conditional survival function is $S_{T_i}(t|T_i > a_i) = S_{T_i}(t)/S_{T_i}(a_i)$.

From these expressions, the log-likelihood from the left-truncated and right-censored survival times can be expressed as

$$\begin{aligned} l(\boldsymbol{\beta}, h_0(t)) &= \sum_{i=1}^n \left\{ \delta_i \log h_{T_i}(y_i) + \log(S_{T_i}(y_i)/S_{T_i}(a_i)) \right\} \\ &= \sum_{i=1}^n \delta_i (\mathbf{x}_i^\top \boldsymbol{\beta} + \log h_0(y_i)) - \sum_{i=1}^n e^{\mathbf{x}_i^\top \boldsymbol{\beta}} H_0(a_i, y_i), \end{aligned} \quad (2)$$

where $H_0(a_i, y_i) = \int_{a_i}^{y_i} h_0(s)ds$, representing the cumulative baseline hazard from a_i to y_i .

Note that the baseline hazard $h_0(t)$ in (1) is an infinite dimensional parameter. A common approach to alleviate this infinite dimensional issue is to approximate $h_0(t)$ using a finite number of basis functions (e.g. Chen et al. (2017); Ma et al. (2021)), where the number of bases can increase with the sample size. This is also known as the method of sieves (Grenander, 1981). Therefore, if the adopted basis functions are $\psi_1(t), \dots, \psi_m(t)$, then an approximation to $h_0(t)$ is

$$h_0(t) = \sum_{u=1}^m \theta_u \psi_u(t), \quad (3)$$

where $\psi_u(t) \geq 0$ for all u , and m may increase with n . In this setting, the problem of estimating $h_0(t)$ becomes a simpler problem of estimating all $\theta_u \geq 0$. Here, the non-negativity constraint on θ_u ensures that $h_0(t) \geq 0$. Another approach, which presents greater computational challenges, is to approximate $\log h_0(t)$ using basis functions (Cai and Betensky, 2003) so that $h_0(t) \geq 0$ is automatically satisfied. The basis functions are typically defined through a sequence of knots, which necessitates the pre-selection of both the number and location of these knots.

Let $\boldsymbol{\theta} = (\theta_1, \dots, \theta_m)^\top$. Estimation of $\boldsymbol{\beta}$ and $\boldsymbol{\theta}$ of the Cox model in (1) can be achieved by maximizing the log-likelihood in (2) subject to all $\theta_u \geq 0$. This constrained optimization can be solved in different ways, such as the Newton-MI algorithm developed in Ma et al. (2021). In this paper, however, we propose a different but more efficient approach by adopting indicator basis functions in (3). In this context, the baseline hazard $h_0(t)$ is approximated by a piecewise constant function. The key advantage of this approach is that it enables the development of a highly efficient algorithm based on the profile likelihood function.

Let the support of $h_0(t)$ be $(a, b]$, where $a = \min_i \{a_i\}$ and $b = \max_i \{y_i\}$. Using a sequence of knots $a = w_1 < w_2 < \dots < w_m < w_{m+1} = b$, we can divide $(a, b]$ into m mutually exclusive and exhaustive bins $\mathcal{B}_1, \dots, \mathcal{B}_m$, where $\mathcal{B}_u = (w_u, w_{u+1}]$ for $u = 1, \dots, m$. Here, $w_1 = a$ and $w_{m+1} = b$ are the boundary knots, and clearly $(a, b] = \bigcup_{u=1}^m \mathcal{B}_u$. There are different approaches to selecting these knots. One is to use equal distance knots, and another is to use quantile-based knots. Usually, quantile-based knots are the preferred choice.

Now we define the following basis functions: $\psi_u(t) = I(t \in \mathcal{B}_u)$, where $I(\cdot)$ is an indicator function such that $\psi_u(t) = 1$ if $t \in \mathcal{B}_u$ and $\psi_u(t) = 0$ otherwise. Based on this partition, we approximate $h_0(t)$ by

$$h_0(t) \approx \sum_{u=1}^m \theta_u \psi_u(t) = \theta_u, \quad \text{if } t \in \mathcal{B}_u,$$

where $u = 1, \dots, m$. From this piecewise constant approximated $h_0(t)$, the $H_0(a_i, y_i)$ function in (2) can be expressed as

$$H_0(a_i, y_i) = \sum_{u=1}^m \theta_u \Delta_{iu} I\{w_u \leq y_i, w_{u+1} \geq a_i\} \quad (4)$$

where

$$\begin{aligned} \Delta_{iu} = & (y_i - a_i) I\{w_u \leq a_i < y_i \leq w_{u+1}\} + (w_{u+1} - a_i) I\{w_u \leq a_i \leq w_{u+1} \leq y_i\} \\ & + (w_{u+1} - w_u) I\{a_i \leq w_u, y_i \geq w_{u+1}\} + (y_i - w_u) I\{a_i \leq w_u \leq y_i \leq w_{u+1}\}. \end{aligned} \quad (5)$$

Therefore, in this context, the log-likelihood in (2) becomes

$$l(\boldsymbol{\beta}, \boldsymbol{\theta}) = \sum_{i=1}^n \delta_i \mathbf{x}_i^\top \boldsymbol{\beta} + \sum_{u=1}^m n_u \log \theta_u - \sum_{i=1}^n e^{\mathbf{x}_i^\top \boldsymbol{\beta}} \sum_{u=1}^m \theta_u \Delta_{iu} I\{w_u \leq y_i, w_{u+1} \geq a_i\},$$

where n_u denotes the number of events that occurred in bin $\mathcal{B}_u$. To obtain the profile log-likelihood, we first find the maximum likelihood solution of $\boldsymbol{\theta}$ as a function of $\boldsymbol{\beta}$. This can be achieved by solving the following gradient equation for $\boldsymbol{\theta}$:

$$\frac{\partial l(\boldsymbol{\beta}, \boldsymbol{\theta})}{\partial \theta_u} = \frac{n_u}{\theta_u} - \sum_{i=1}^n e^{\mathbf{x}_i^\top \boldsymbol{\beta}} \Delta_{iu} I\{w_u \leq y_i, w_{u+1} \geq a_i\} = 0.$$

Its solution can be expressed in a closed form as

$$\theta_u = \frac{n_u}{\sum_{i=1}^n e^{\mathbf{x}_i^\top \boldsymbol{\beta}} \Delta_{iu} I\{w_u \leq y_i, w_{u+1} \geq a_i\}}. \quad (6)$$

We comment that:

- (i) The solution for θ_u provided in (6) always satisfies $\theta_u \geq 0$, where equality holds only when $n_u = 0$. Thus, to avoid any θ_u estimate becomes zero, we recommend that each $\mathcal{B}_u$ contains at least one event time.
- (ii) The summation in the denominator in (6) clarifies that an individual i contributes to the calculation of θ_u (associated with interval $\mathcal{B}_u$) only if (a) their observation is greater than w_u , the left limit of $\mathcal{B}_u$, and (b) their left truncation time is located before w_{u+1} , the right limit of $\mathcal{B}_u$. These conditions define a risk set for estimating $\boldsymbol{\beta}$ using the profile log-likelihood, with further discussions given below.

By substituting the solutions for all θ_u back into the log-likelihood $l(\boldsymbol{\beta}, \boldsymbol{\theta})$, we obtain the so-called profile log-likelihood, which is a function of $\boldsymbol{\beta}$ only. This profile log-likelihood can be written as

$$l_p^*(\boldsymbol{\beta}) \propto \sum_{i=1}^n \delta_i \mathbf{x}_i^\top \boldsymbol{\beta} - \sum_{u=1}^m n_u \log \left(\sum_{i=1}^n e^{\mathbf{x}_i^\top \boldsymbol{\beta}} \Delta_{iu} I\{w_u \leq y_i, w_{u+1} \geq a_i\} \right), \quad (7)$$

where the terms independent of β are ignored.

A foreseeable advantage of the profile likelihood approach is its fast computation speed, as a Newton algorithm can be easily implemented to maximize the profile log-likelihood (7).

3 Maximum profile log-likelihood estimation

Using the profile log-likelihood, we can estimate β by

$$\hat{\beta} = \underset{\beta}{\operatorname{argmax}} l_p^*(\beta). \quad (8)$$

To achieve this, the Newton algorithm can be used to calculate $\hat{\beta}$ iteratively through

$$\beta^{(k+1)} = \beta^{(k)} - \left[\frac{\partial^2 l_p^*(\beta^{(k)})}{\partial \beta \partial \beta^\top} \right]^{-1} \frac{\partial l_p^*(\beta^{(k)})}{\partial \beta}. \quad (9)$$

To implement this algorithm, we need to work out the gradient $\partial l_p^*(\beta^{(k)})/\partial \beta$ and the Hessian matrix $\partial^2 l_p^*(\beta^{(k)})/\partial \beta \partial \beta^\top$.

Let $\tilde{w}_{iu}$ be:

$$\tilde{w}_{iu} = \frac{e^{\mathbf{x}_i^\top \beta} \Delta_{iu} I\{w_u \leq t_i^*, w_{u+1} \geq L_i\}}{\sum_{i=1}^n e^{\mathbf{x}_i^\top \beta} \Delta_{iu} I\{w_u \leq t_i^*, w_{u+1} \geq L_i\}}. \quad (10)$$

As $\tilde{w}_{iu} \geq 0$ and $\sum_i \tilde{w}_{iu} = 1$, we can view each $\tilde{w}_{iu}$ as a weight assigned from bin $\mathcal{B}_u$ to individual i . Let $\mathbf{X} = (\mathbf{x}_1, \dots, \mathbf{x}_n)^\top$, $\boldsymbol{\delta} = (\delta_1, \dots, \delta_n)^\top$, $\widetilde{\mathbf{W}} = (\tilde{w}_{iu})_{n \times m}$ and $\mathbf{n}_m = (n_1, \dots, n_m)^\top$. Now, the gradient vector can be expressed in a matrix form as:

$$\frac{\partial l_p^*}{\partial \beta} = \mathbf{X}^\top (\boldsymbol{\delta} - \widetilde{\mathbf{W}} \mathbf{n}_m). \quad (11)$$

The Hessian matrix is given by

$$\frac{\partial^2 l_p^*}{\partial \beta \partial \beta^\top} = -\mathbf{X}^\top \operatorname{diag} \left(\sum_{u=1}^m n_u \widetilde{\mathbf{W}}_{[u]} \right) \mathbf{X} - \overline{\mathbf{X}} \operatorname{diag}(\mathbf{n}_m) \overline{\mathbf{X}}^\top, \quad (12)$$

where $\widetilde{\mathbf{W}}_{[u]}$ denotes the u -th column of matrix $\widetilde{\mathbf{W}}$ and $\overline{\mathbf{X}} = \mathbf{X}^\top \widetilde{\mathbf{W}}$. Note that the dimension of $\overline{\mathbf{X}}$ is $p \times m$.

The gradient and Hessian formulas given in (11) and (12) make the Newton algorithm extremely easy to implement.

Remark on related implementations. A related implementation of piecewise constant hazard models for survival data is available in the R package `eha`, through the function `pchreg()` (Broström, 2022). More recently, a similar routine has been distributed as the separate R package `pch` (Harrell, 2022). Both implementations allow for the estimation of Cox-type models under left truncation and right censoring using

a parametric piecewise constant baseline hazard. However, the methodological details underlying these implementations, including the derivation of the likelihood and the estimation procedures, have not been formally published. Our proposed approach shares conceptual similarities with these packages but differs in key computational aspects: we employ a dedicated efficient Newton algorithm to maximize the profile log-likelihood, while the existing implementations rely on the general-purpose `optim()` function. Moreover, our work explicitly derives the full likelihood formulation, provides theoretical justification, and establishes asymptotic results, thereby formalizing and extending the methodology underlying these earlier software implementations.

4 Asymptotic properties

As explained in Murphy and van der Vaart (2000), a profile likelihood can be used as a conventional likelihood, and it is common practice to employ the curvature of the profile likelihood function to estimate the variability of the estimated parameters. In fact, under some specific regularity conditions, the distribution of $\sqrt{n}(\hat{\beta} - \beta_0)$, where β_0 represents the true coefficient vector and $\hat{\beta}$ is the maximum profile likelihood estimator, converges when $n \rightarrow \infty$ to a multivariate normal distribution with a mean vector $\mathbf{0}$ and a covariance matrix of $-[n^{-1}\partial^2 l_p^*(\hat{\beta})/\partial\beta\partial\beta^\top]^{-1}$. This asymptotic outcome closely resembles that of the traditional Cox partial likelihood and suffices if one's interest lies solely in inferences about β .

However, for some applications, such as when the model is used for prediction, the variance of not only β but also θ are demanded. In this context, inferences call for the asymptotic distribution involving both β and θ , where the dimension of θ is preselected and thus fixed for a given dataset. Let $\eta = (\beta^\top, \theta^\top)^\top$. Then under certain regularity conditions and when $n \rightarrow \infty$, the distribution of $\sqrt{n}(\hat{\eta} - \eta_0)$ converges to $N(\mathbf{0}, n\mathbf{F}^{-1})$, where η_0 represents the true parameter vector and $\mathbf{F}$ is the negative Hessian matrix of the log-likelihood $l(\eta)$ at η_0 . Since η_0 is usually unavailable, we can replace it with the maximum likelihood estimate $\hat{\eta}$.

Computation of $\mathbf{F}^{-1}$ is generally straightforward. However, when the dimension of θ (i.e., m) is large, it is more efficient to invert $\mathbf{F}$ utilizing the fact that $\partial^2 l/\partial\theta\partial\theta^\top$ is a diagonal matrix. To the end, we partition $\mathbf{F}$ into a 2×2 block matrix:

$$\mathbf{F} = \begin{bmatrix} \mathbf{F}_{11} & \mathbf{F}_{12} \\ \mathbf{F}_{12}^\top & \mathbf{F}_{22} \end{bmatrix},$$

where $\mathbf{F}_{11} = -\partial^2 l/\partial\beta\partial\beta^\top$, $\mathbf{F}_{12} = -\partial^2 l/\partial\beta\partial\theta^\top$ and $\mathbf{F}_{22} = -\partial^2 l/\partial\theta\partial\theta^\top$. The elements of $\mathbf{F}_{11}$ are

$$\sum_{i=1}^n x_{ij}x_{it}e^{\mathbf{x}_i^\top\beta} \sum_{u=1}^m \theta_u \Delta_{iu} I\{w_u \leq y_i, w_{u+1} \geq a_i\},$$

and the elements of $\mathbf{F}_{12}$ are

$$\sum_{i=1}^n x_{ij} e^{\mathbf{x}_i^\top \boldsymbol{\beta}} \Delta_{iu} I\{w_u \leq y_i, w_{u+1} \geq a_i\}.$$

Finally, $\mathbf{F}_{22}$ is a diagonal matrix with diagonal elements n_u/θ_u^2 . Taking advantage of the fact that $\mathbf{F}_{22}$ is diagonal, $\mathbf{F}^{-1}$ can be efficiently calculated according to

$$\mathbf{F}^{-1} = \begin{bmatrix} (\mathbf{F}_{11} - \mathbf{F}_{12}\mathbf{F}_{22}^{-1}\mathbf{F}_{12})^{-1} & -(\mathbf{F}_{11} - \mathbf{F}_{12}\mathbf{F}_{22}^{-1}\mathbf{F}_{12})^{-1}\mathbf{F}_{12}\mathbf{F}_{22}^{-1} \\ -\mathbf{F}_{22}^{-1}\mathbf{F}_{12}(\mathbf{F}_{11} - \mathbf{F}_{12}\mathbf{F}_{22}^{-1}\mathbf{F}_{12})^{-1} & \mathbf{F}_{22}^{-1} + \mathbf{F}_{22}^{-1}\mathbf{F}_{12}(\mathbf{F}_{11} - \mathbf{F}_{12}\mathbf{F}_{22}^{-1}\mathbf{F}_{12})^{-1}\mathbf{F}_{12}\mathbf{F}_{22}^{-1} \end{bmatrix}. \quad (13)$$

5 Simulation Results

We conducted a comprehensive simulation study to evaluate the performance of our proposed method relative to the Cox proportional hazards (CoxPH) model implemented in the `coxph()` function from the R `survival` package, which allows for both left truncation and right censoring.

The objectives of this simulation study were twofold: (i) to evaluate the accuracy and reliability of the regression coefficient estimates from both methods under varying censoring schemes and sample sizes, and (ii) to assess the ability of the profile likelihood approach to recover baseline hazard information. The profile likelihood approach has a distinct advantage in this regard, as it provides both baseline hazard estimates and pointwise confidence intervals. In contrast, the `coxph()` function does not directly estimate the baseline hazard.

We now briefly describe the data generating mechanism used. The event times were generated from a Weibull distribution with the baseline hazard as $h_0(t) = 3t^2$, and with two independent covariates: $X_1 \sim \text{Bernoulli}(0.5)$ and $X_2 \sim \mathcal{N}(0, 1)$. The event times were then obtained using inverse transform sampling. Left truncation and right censoring times were sampled independently from uniform distributions to emulate realistic incomplete follow-up.

We considered sample sizes of $n = 200$ and $n = 1000$ to represent small and large samples, each under moderate (30%) and heavy (70%) right-censoring. In order to examine the effect of level of discretization on the piecewise hazard approximation, let m denote the number of bins. For $n = 200$, we explored $m \in \{5, 10, 20\}$; for $n = 1000$, we considered $m \in \{10, 20, 30\}$. These settings balance practical relevance with computational feasibility while covering a range of discretization levels.

Performance was quantified using bias, Monte Carlo standard deviation (MCSD), average asymptotic standard deviation (AASD) and coverage probabilities (evaluated using both MCSD- and AASD-based standard errors). The results from this simulation

Table 1: Simulation results for $n = 200$, varying m . The CoxPH values are constant across m for each metric.

m	Metric	30% Right-Censoring				70% Right-Censoring			
		Profile		CoxPH		Profile		CoxPH	
		β_1	β_2	β_1	β_2	β_1	β_2	β_1	β_2
5	Bias	-0.054	0.040			0.007	-0.014		
10		-0.005	-0.012	0.005	-0.025	0.029	-0.038	0.020	-0.031
20		0.008	-0.027			0.034	-0.044		
5	MCSD	0.174	0.114			0.284	0.161		
10		0.181	0.124	0.181	0.128	0.290	0.168	0.286	0.167
20		0.183	0.127			0.290	0.171		
5	AASD	0.183	0.112			0.276	0.159		
10		0.186	0.116	0.187	0.118	0.277	0.162	0.278	0.163
20		0.186	0.117			0.278	0.163		
5	CP(MCSD)	0.943	0.943			0.943	0.927		
10		0.950	0.943	0.937	0.950	0.940	0.927	0.947	0.930
20		0.940	0.950			0.933	0.927		
5	CP(AASD)	0.953	0.927			0.940	0.920		
10		0.953	0.930	0.953	0.923	0.930	0.920	0.940	0.923
20		0.950	0.933			0.927	0.917		

study, which evaluate the accuracy of the parameter estimates, are summarized in Tables 1 and 2.

For $n = 200$, both the profile likelihood and CoxPH methods produced small biases. Although minor variations in bias were observed, the variability of the profile likelihood estimates was largely insensitive to the choice of m . Moreover, MCSD and AASD values from the profile likelihood method closely matched those from CoxPH.

However, under these two censoring schemes, both methods exhibit slight under-coverage for β_2 , indicating a mild underestimation of variability by the Monte Carlo and asymptotic standard errors. Such underestimation can occur when information is unevenly distributed across risk sets, for example, when a Weibull hazard causes events to cluster toward early or late times, reducing the effective sample size in certain regions and creating small discrepancies between asymptotic approximations and empirical variability.

For the large-sample setting ($n = 1000$), the profile likelihood method performed essentially on par with the CoxPH model for β_1 and β_2 under both moderate (30%) and heavy (70%) censoring, with little variability across various values of m . Coverage probabilities for both methods were near the nominal 95% level across most scenarios,

Table 2: Simulation results for $n = 1000$, varying m . The CoxPH values are constant across m for each metric.

m	Metric	30% Right-Censoring				70% Right-Censoring			
		Profile		CoxPH		Profile		CoxPH	
		β_1	β_2	β_1	β_2	β_1	β_2	β_1	β_2
10	Bias	-0.024	0.015			-0.008	0.001		
20		-0.007	-0.004	-0.002	-0.010	-0.003	-0.006	-0.003	-0.005
30		-0.003	-0.008			-0.001	-0.007		
10	MCSD	0.086	0.055			0.123	0.072		
20		0.088	0.056	0.089	0.057	0.124	0.073	0.124	0.073
30		0.088	0.057			0.124	0.073		
10	AASD	0.081	0.050			0.120	0.069		
20		0.082	0.051	0.082	0.051	0.120	0.070	0.121	0.070
30		0.082	0.051			0.120	0.070		
10	CP(MCSD)	0.940	0.947			0.947	0.950		
20		0.953	0.960	0.953	0.950	0.953	0.953	0.953	0.953
30		0.950	0.953			0.953	0.950		
10	CP(AASD)	0.940	0.913			0.947	0.940		
20		0.943	0.930	0.943	0.933	0.947	0.940	0.950	0.943
30		0.943	0.927			0.947	0.940		

with a mild undercoverage with the AASD approach for β_2 under 30% right-censoring, showing a major improvement over the small sample case. Overall, these results confirm that, in large samples, the profile likelihood approach is highly comparable to the CoxPH approach.

While yielding regression estimates comparable to CoxPH, the profile-likelihood method additionally provides baseline hazard estimates and pointwise confidence intervals, making it particularly advantageous when baseline hazard inference is of interest.

Therefore, we computed the estimated piecewise constant baseline hazard and baseline survival from the profile-likelihood and compared them with the theoretical baseline hazard $h_0(t) = 3t^2$ and the corresponding true baseline survival $S_0(t) = \exp(-t^3)$. For visual assessment, we overlaid pointwise 95% confidence intervals computed from both MCSD- and AASD-based standard errors. The corresponding standard deviations for the survival curve values were obtained via the delta method.

Figures 1 and 2 present the pointwise estimated baseline hazard and baseline survival functions, based on our proposed approach, for multiple combinations of sample sizes, censoring proportions, and numbers of bins. The estimated baseline functions

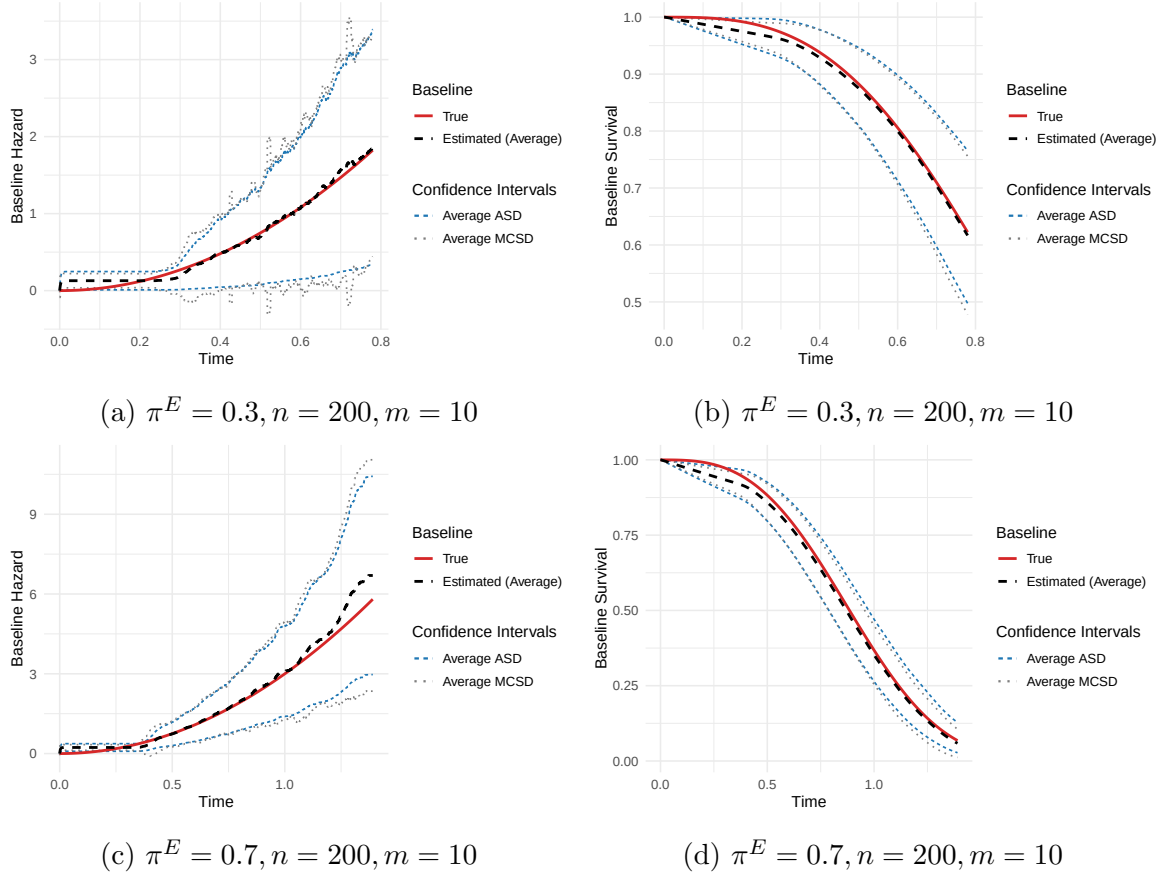


Figure 1: Estimated baseline hazard and survival functions when $n = 200$ and $m = 10$ using the profile likelihood approach. Each plot includes the true baseline hazard or survival curve (solid red line), the mean estimated baseline hazard or survival curve (long-dashed black line), the average 95% pointwise confidence intervals obtained using Monte Carlo standard errors (MCSD) (grey dotted lines), and the average 95% pointwise confidence intervals obtained using asymptotic standard errors (ASD) (blue dashed lines).

track the true baseline functions closely, with accuracy improving over a broader range of t as either the proportion of exact failure times or the sample size increases. Furthermore, the widths of the pointwise MCSD and AASD confidence intervals decrease markedly with larger sample sizes, indicating improved precision of the estimates.

Taken together, these results demonstrate that the profile likelihood approach not only performs on par with the CoxPH model for regression coefficient estimation, but also delivers the important additional benefit of directly estimating the baseline hazard and survival functions, complete with pointwise confidence intervals, which are capabilities that the standard CoxPH model does not provide.

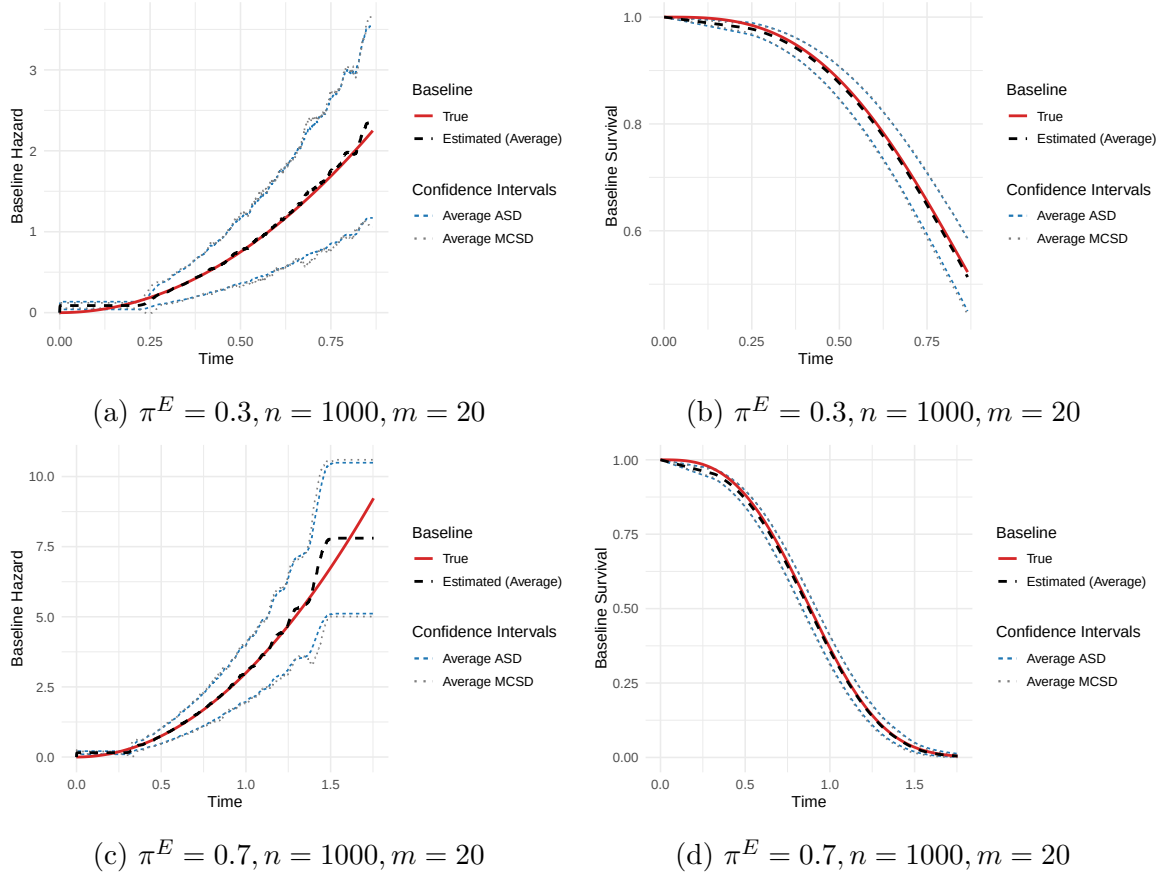


Figure 2: Estimated baseline hazard and survival functions when $n = 1000$ and $m = 20$ using the profile likelihood approach. Each plot includes the true baseline hazard or survival curve (solid red line), the mean estimated baseline hazard or survival curve (long-dashed black line), the average 95% pointwise confidence intervals obtained using Monte Carlo standard errors (MCSD) (grey dotted lines), and the average 95% pointwise confidence intervals obtained using asymptotic standard errors (ASD) (blue dashed lines).

6 Real Data Results

To illustrate the maximum profile likelihood method for Cox models, we apply it to a real-world dataset comprising 1,605 patients diagnosed with multiple primary melanomas. Eligibility required patients to have at least two primary melanoma diagnoses, with the first and second occurring at least 30 days apart. Consequently, the dataset is subject to left truncation.

In this application, we analyze the time to death from melanoma (measured in years). Individuals with a history of melanoma face an elevated risk of developing a second primary melanoma; however, not all patients live long enough to experience this, as some die from the disease beforehand. This induces a selection bias due to left truncation, such that applying a standard Cox proportional hazards model would

result in overestimation of survival time.

Out of the total cohort, 1,341 out of 1,605 patients (83%) have right-censored failure times. The following four covariates are considered in the analysis: age (in years), gender (male or female), Breslow thickness of the first primary melanoma (in millimeters), and SLNBx1, which indicates whether or not a sentinel lymph node biopsy was performed after the first primary melanoma diagnosis.

We then implemented the proposed method with a piecewise approximation to the baseline hazard consisting of 16 bins, determined as the square root of the number of exact event times in the dataset. The optimization process was halted once the changes in the estimates of the model parameters (all the regression coefficients) were within a tolerance level of 10^{-10} .

The estimated regression coefficients and their corresponding standard errors are given in Table 3. Under the asymptotic normality of the estimators, we include Wald z -statistics and p-values. To benchmark our approach, we additionally fit a CoxPH model that incorporates left truncation and right-censoring, with its results appearing in Table 4.

Table 3: Estimation results using the profile likelihood approach.

	$\hat{\beta}$	$se(\hat{\beta})$	Wald statistic	P-value
Age (in years)	0.0457	0.0060	7.605	$2.84e - 14$
Gender (male vs female)	-0.2861	0.1514	-1.890	0.0587
Breslow Thickness (Baseline, in mm)	0.1158	0.0223	5.199	$2.00e - 07$
SLNBx1	0.0039	0.1326	0.0294	0.9766

Table 4: Estimation results using the CoxPH (with left truncation and right censoring) approach.

	$\hat{\beta}$	$se(\hat{\beta})$	Wald statistic	P-value
Age (in years)	0.0460	0.0060	7.630	$2.34e - 14$
Gender (male vs female)	-0.2780	0.1515	-1.834	0.067
Breslow Thickness (Baseline, in mm)	0.1143	0.0222	5.158	$2.50e - 07$
SLNBx1	0.0090	0.1327	0.068	0.9458

Both approaches provide very similar results for the regression coefficients. From the tables, for a 5% significance level, it is evident that age and breslow thickness are statistically significant.

Since the `coxph()` function does not directly estimate the baseline hazard, we obtained the cumulative baseline hazard for the fitted CoxPH model using Breslow's

estimator, and then obtained the baseline survival by transformation. The resulting hazard and survival curves from the CoxPH and profile-likelihood methods are shown in Figures 3 and 4.

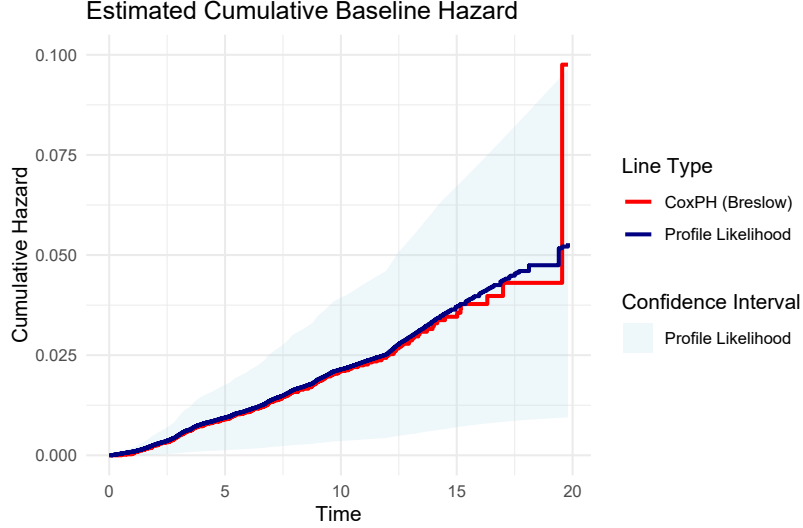


Figure 3: Estimated cumulative baseline hazard using the profile likelihood approach and CoxPH approach on the melanoma dataset. The baseline cumulative hazard estimated from the CoxPH results is shown in red while the estimated baseline from the profile likelihood approach is shown in navy. The region covered by the 95% pointwise confidence intervals obtained from the profile likelihood approach is shaded in light blue.

Overall, the trajectory of the cumulative baseline hazard $\hat{H}_0(t)$ estimated by the profile-likelihood method closely tracks the Breslow estimate from the fitted CoxPH model, and the corresponding baseline survival curves $\hat{S}_0(t)$ likewise show close agreement. A further advantage of the profile-likelihood approach is that it yields valid pointwise confidence intervals for $H_0(t)$ and for $S_0(t) = \exp\{-H_0(t)\}$ via the delta method, enabling formal inference on both scales. Small discrepancies at the boundaries reflect reduced information in the risk sets, where Breslow’s estimator is known to be less stable.

7 Conclusion

This paper considers a novel profile likelihood-based fast estimation method for fitting Cox models under left truncation and right censoring. Compared with the conventional partial likelihood method for this type of data, the proposed method achieves a comparable rate of convergence. The simulation study reported in this paper demonstrates that the two methods generally produce very similar regression coefficient estimates in

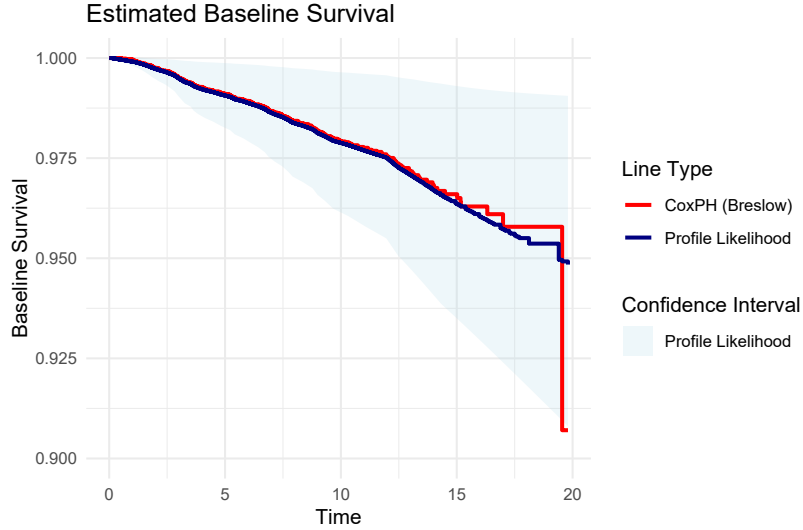


Figure 4: Estimated baseline survival using the profile likelihood approach and CoxPH approach on the melanoma dataset. The baseline survival estimated from the CoxPH results is shown in red while the estimated baseline from the profile likelihood approach is shown in navy. The region covered by the 95% pointwise confidence intervals obtained from the profile likelihood approach is shaded in light blue.

terms of both accuracy and standard deviation. However, unlike the partial likelihood approach, the proposed method simultaneously provides an estimate of the baseline hazard, approximated by a piecewise constant function. This is an advantage over the partial likelihood method in practice, as our method enables inference for quantities (such as individual survival functions) that involve both the regression coefficients and the baseline hazard.

We are currently working on extending our method to accommodate more general Cox models that allow for left truncation, interval censoring, and time-dependent covariates.

Declarations

Abbreviations

AASD - Average Asymptotic Standard Deviation

ASD - Asymptotic Standard Deviation

CoxPH - Cox Proportional Hazard

MCSD - Monte Carlo Standard Deviation

Ethics Approval and Consent to Participate

Not applicable.

Consent for Publication

Not applicable.

Availability of Data and Materials

The dataset may be accessed upon reasonable request from the Melanoma Institute Australia.

Competing Interests

Not applicable.

Clinical Trial Number

Not applicable.

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Author's Contributions

J.M. developed the methodology, while A.B. and B.L. helped review and refine it. A.B. developed the code. All authors contributed to writing and reviewing the manuscript.

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